

Black tea is not significantly different from water in the maintenance of normal hydration in human subjects: results from a randomised controlled ...



There is a belief that caffeinated drinks, such as tea, may adversely affect hydration. This was investigated in a randomised controlled trial. Healthy resting males (n 21) were recruited from the general population. Following 24 h of abstention from caffeine, alcohol and vigorous physical activity, including a 10 h overnight fast, all men underwent four separate test days in a counter-balanced order with a 5 d washout in between. The test beverages, provided at regular intervals, were 4 × 240 ml black (i.e. regular) tea and 6 × 240 ml black tea, providing 168 or 252 mg of caffeine. The controls were identical amounts of boiled water. The tea was prepared in a standardised way from tea bags and included 20 ml of semi-skimmed milk. All food taken during the 12 h intervention period was controlled, and subjects remained at rest. No other beverages were offered. Blood was sampled at 0, 1, 2, 4, 8 and 12 h, and a 24 h urine sample was collected. Outcome variables were whole blood cell count, Na, K, bicarbonate, total protein, urea, creatinine and osmolality for blood; and total volume, colour, Na, K, creatinine and osmolality for urine. Although data for all twenty-one participants were included in the analysis (mean age 36 years and mean BMI 25·8 kg/m²), nineteen men completed all conditions. Statistical analysis, using a factorial ANOVA approach within PROC MIXED, revealed no significant differences between tea and water for any of the mean blood or urine measurements. It was concluded that black tea, in the amounts studied, offered similar hydrating properties to water.